Phonological Rules & Derivation

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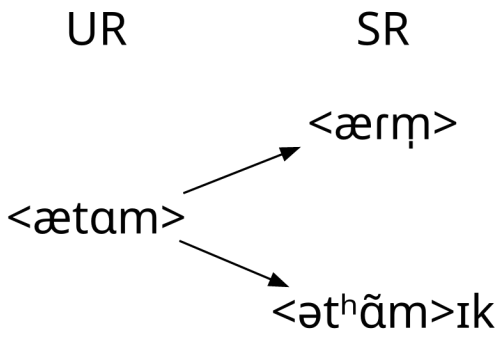
Underlying representations

A single morpheme might be pronounced multiple ways

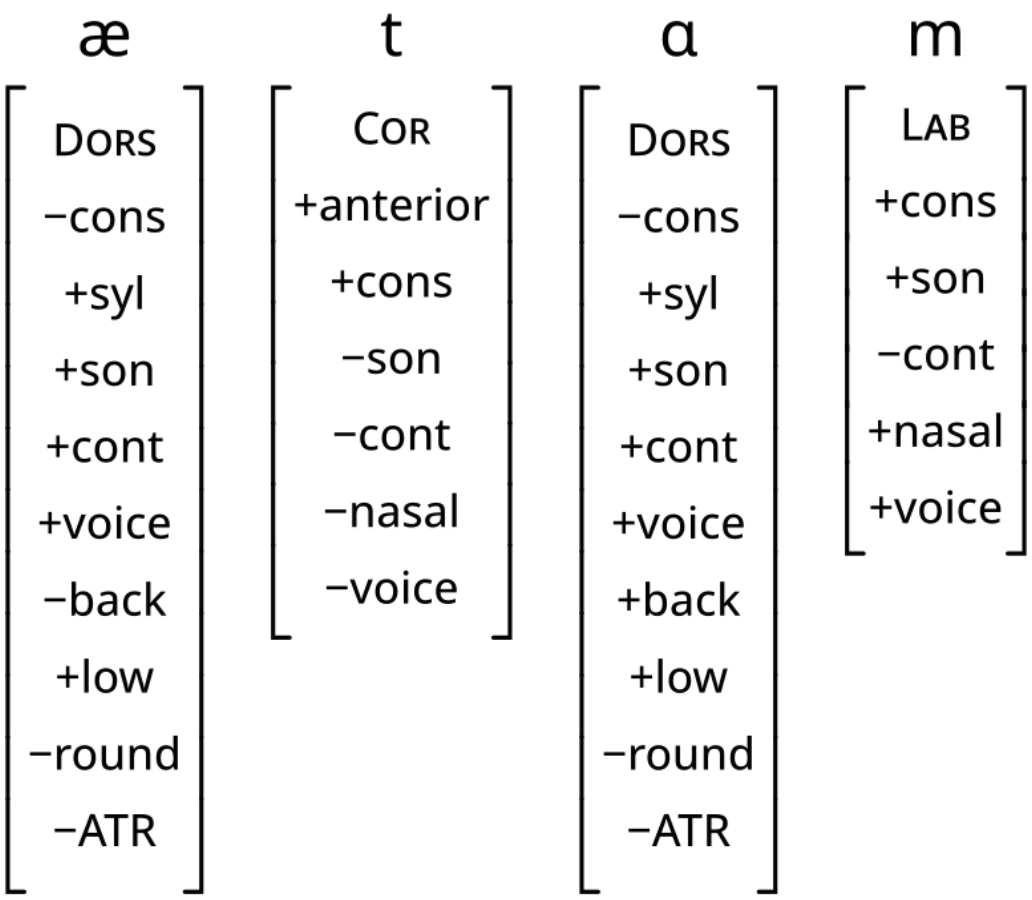
“atom”	<ærm̩>
“atomic”	<ətʰm̃>ɪk

Underlying representations

The phonological hypothesis is that we represent a single “underlying representation” of the word in our mental lexicon, which then gets transformed depending on context.

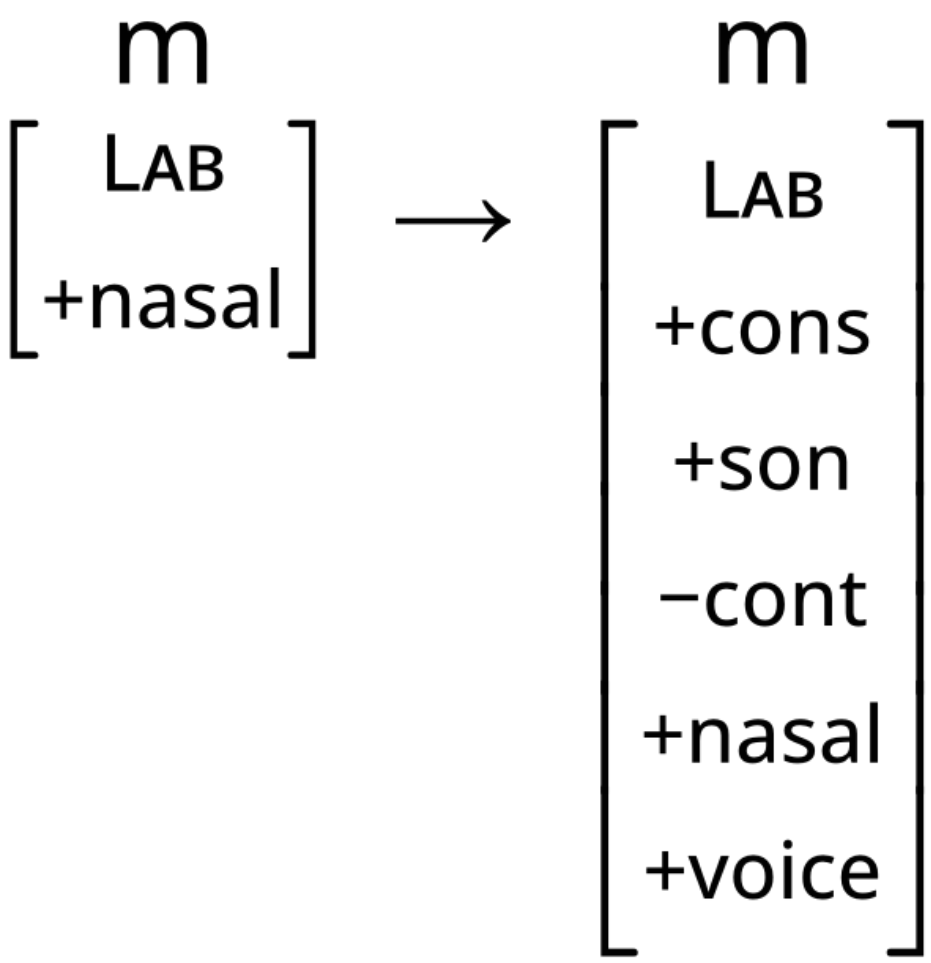


Underlying representations



Underlying representations

"Contrastive specification"



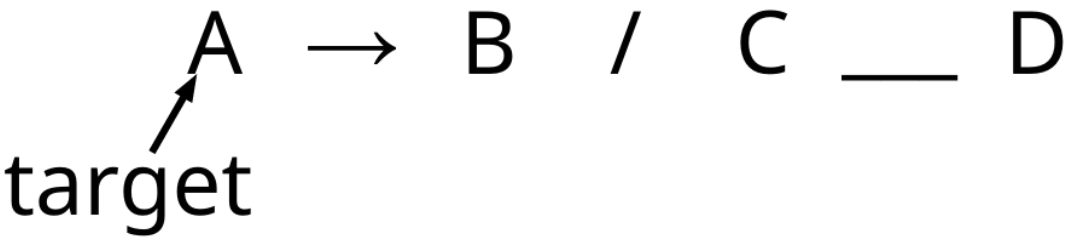
Phonological Rules

Anatomy of a rule

$$A \rightarrow B \;/\; C ___ D$$

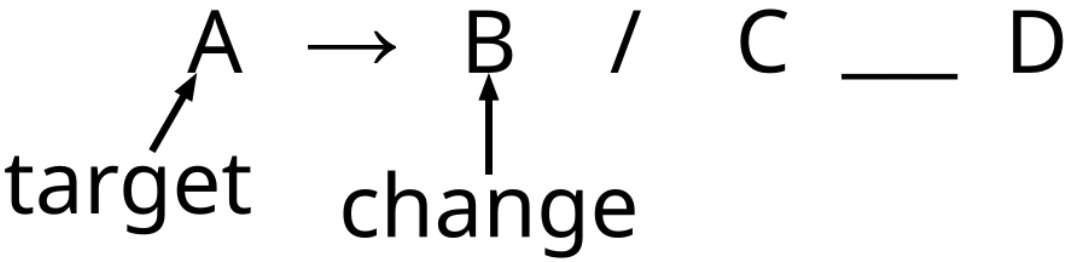
Phonological Rules

Anatomy of a rule



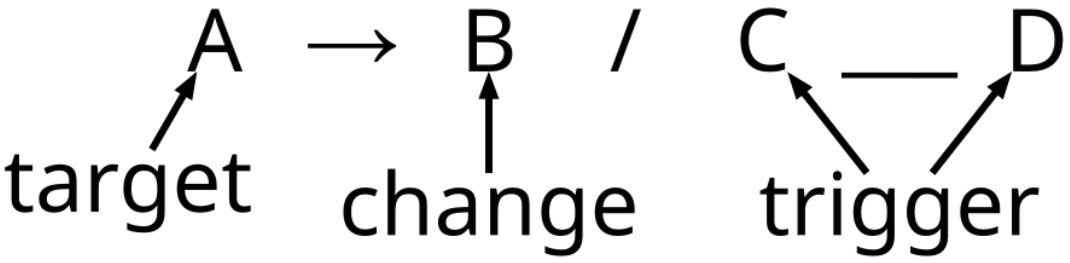
Phonological Rules

Anatomy of a rule



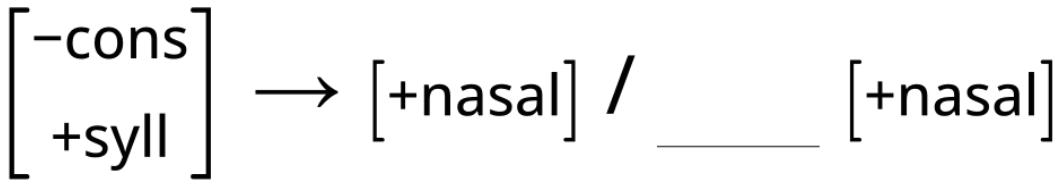
Phonological Rules

Anatomy of a rule



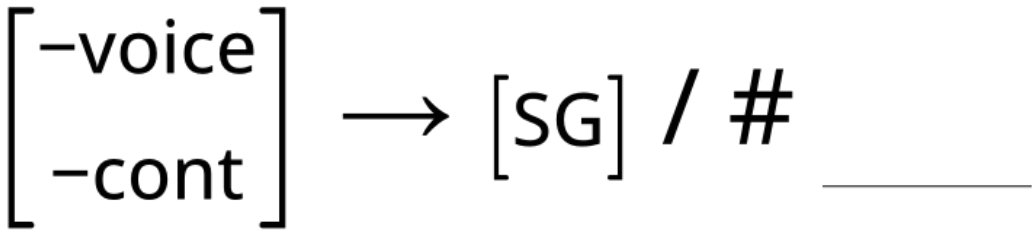
Phonological Rules

“Vowels become nasal before a nasal phoneme”



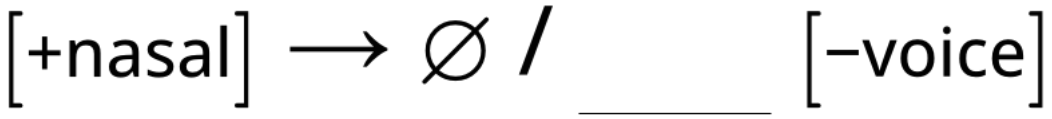
Phonological Rules

“Voiceless stops become aspirated at the beginning of words”



Phonological Rules

“A nasal stop gets deleted before a voiceless stop”



Phonologial Rules

“A schwa gets inserted between two coronal strident fricatives”



Phonological Rules

"A vowel becomes round if the preceding vowel is round, even if a consonant intervenes"

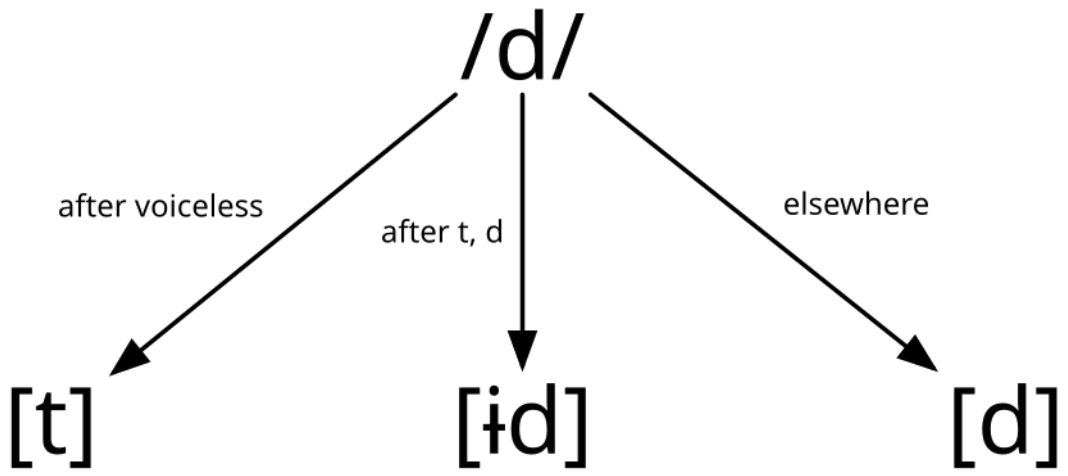
$$V \longrightarrow [+round] \; / \; \underset{[+round]}{V} \; (C) \; ____$$

Phonological rules

Let's remember our English Past tense data

snowed	sno-d	*faced*	fes-t	*waited*	wet-ɨd
banned	bæn-d	*joked*	dʒok-t	*punted*	pʌnt-ɨd
called	kɔl-d	*miffed*	mɪf-t	*raided*	ɹed-ɨd
posed	poz-d	*capped*	kæp-t	*beaded*	bid-ɨd

Phonological Rules



Phonological rules

$$\textsc{Assimilation}: [-\text{son}] \longrightarrow [-\text{voice}] / [-\text{voice}] ___$$

$$\textsc{Epenthesis}: \varnothing \longrightarrow \begin{bmatrix} +\text{syll} \\ +\text{hi} \\ +\text{back} \\ -\text{round} \end{bmatrix} / \begin{bmatrix} \textsc{Cor} \\ -\text{cont} \\ -\text{son} \end{bmatrix} ___ \begin{bmatrix} \textsc{Cor} \\ -\text{cont} \\ -\text{son} \end{bmatrix}$$

Phonological Rules

What *order* do they go in?

UR	sno+d	mɪs+d	lɪft+d	UR	sno+d	mɪs+d	lɪft+d
Assimilation				Epenthesis			
Epenthesis				Assimilation			
SR				SR			

Phonological Rules

UR	sno+d	mɪs+d	lɪft+d
Assimilation	–	mɪst	lɪftt
Epenthesis	–	–	lɪftɨt
SR	snod	mɪst	*lɪftɨt

Phonological Rules

UR	sno+d	mɪs+d	lɪft+d
Epenthesis	–	–	lɪftɨd
Assimilation	–	mɪst	–
SR	snod	mɪst	lɪftɨd

Phonological Rules

We would say that Epenthesis and Assimilation are "Crucially Ordered." We have names for some kinds of crucial ordering

- "Bleeding"
- "Feeding"

Phonological Rules

UR	lɪft+d	*The context exists for Assimilation to apply.*
Epenthesis	lɪftɨd	*The context for Assimilation is eliminated.*
Assimilation	–	*Assimilation does not apply.*
SR	lɪftɨd	

Epenthesis "bleeds" Assimilation

Phonological Rules

$$\textsc{Flapping:}\ \begin{bmatrix} \textsc{Cor} \\ -\text{cont} \\ -\text{son} \end{bmatrix} \longrightarrow [+\text{son}] \ /\ V ____ \begin{bmatrix} +\text{syll} \\ -\text{stress} \end{bmatrix}$$

Phonological Rules

UR	wet+d	*The context for Flapping does not exist.*
Epenthesis	wetɨd	*The context for Flapping is created.*
Assimilation	–	
Flapping	weɾɨd	*Flapping applies.*
SR	weɾɨd	

Epenthesis "feeds" Flapping

Phonological Rules

UR	wetd	bʌtɻ
Flapping	–	bʌɾɻ
Epenthesis	wetɨd	–
SR	*wetɨd	bʌɾɻ

UR	wetd	bʌtɻ
Epenthesis	wetɨd	–
Flapping	weɾɨd	bʌɾɻ
SR	weɾɨd	bʌɾɻ

Rule interaction

Rule Interaction Handout